QUESTION 32: EXPLAIN MARINE POLLUTION MONITORING USING MACHINE LEARNING

Related Sustainable Development Goal: SDG 14 – Life Below Water

1. Introduction

Marine pollution is the entry of harmful materials into seas and oceans. These materials include plastic waste, oil, untreated sewage, industrial chemicals, agricultural run-off and other pollutants. Pollution can damage coral reefs, harm fish and marine mammals, affect seafood safety and disturb the balance of marine ecosystems. It can also affect coastal communities whose income and food supply depend on the ocean.

Monitoring pollution is difficult because the ocean is very large, conditions change with tides and weather, and pollution may spread over wide areas. Traditional monitoring often depends on collecting water samples and testing them in a laboratory. This method is important, but it can take time, cover only a limited number of locations and miss a sudden event between sampling visits.

Machine Learning (ML) can support monitoring by learning patterns from previous and current observations. It can combine measurements from water-quality sensors, satellite images, drones, research vessels and verified incident reports to identify suspicious conditions, estimate risk and highlight places that need inspection. ML does not replace laboratory testing or environmental experts; it helps them decide where and when to investigate.

This assignment presents a practical ML-based monitoring approach and discusses the problem, data, preparation, algorithms, workflow, evaluation, visualization, limitations and recommendations.

2. Problem Statement and Objectives

Problem statement: Coastal and marine authorities need a timely way to identify locations where pollution may be present, classify the likely pollution level and prioritize areas for further testing. A Machine Learning system can analyse environmental observations and flag suspicious patterns for human review.

Objectives

- Use environmental and location-based data to identify possible marine pollution.
- Classify observations into low, moderate and high pollution-risk categories.
- Support early warning and help monitoring teams plan sampling and cleanup activities.
- Present results through understandable charts, maps or a monitoring dashboard.
- Connect the proposed solution to SDG 14, especially reducing marine pollution and protecting ocean ecosystems.

3. Background Study and SDG 14 Linkage

Marine pollution comes from both land-based and sea-based sources. Land-based sources include plastic litter carried by rivers, sewage discharge, fertilizers, pesticides and industrial waste. Sea-based sources include ship operations, accidental oil spills, abandoned fishing gear and offshore activities.

Pollutants do not always remain near their source: currents, wind, rainfall and tides can move them to other places.

Different forms of pollution require different measurements. Turbidity and suspended solids may indicate sediment or waste entering the water. Dissolved oxygen, pH, temperature and conductivity describe water conditions. Satellite images can help identify large surface features, floating debris accumulations or oil-like slicks under suitable conditions. However, no single feature proves pollution is present. Turbidity can rise naturally after a storm, so the model must consider context and uncertainty.

The relevant global goal is SDG 14 – Life Below Water. Target 14.1 calls for preventing and significantly reducing marine pollution, particularly pollution from land-based activities, including marine debris and nutrient pollution. An ML monitoring system can contribute by making monitoring more frequent, directing attention to possible hotspots and helping authorities respond sooner. It is a decision-support tool, not a substitute for pollution prevention policies or enforcement.

The social value is practical: earlier identification may help protect fisheries, coastal tourism, biodiversity and public health. The system is most useful when predictions lead to follow-up sampling, source investigation and action to reduce pollution.

4. Defining the Machine Learning Problem

4.1 Learning task

The main task can be treated as supervised classification. Each historical observation is labelled using reliable water testing, expert inspection or an official incident record. The model learns the relationship between observed features and known pollution categories. After training, it estimates a category for new observations.

4.2 Input features and target output

Type	Example variables	Purpose
Water measurements	Turbidity, dissolved oxygen, pH, temperature, conductivity	Describe physical and chemical water conditions.
Location and time	Latitude, longitude, date, season, distance from shore	Represent where and when the sample was collected.
Weather and ocean context	Rainfall, wind speed, waves, tide or current information	Help distinguish pollution signals from natural changes.
Remote-sensing features	Surface-colour indicators, image texture, slick-like patterns	Provide wider-area observations where imagery is suitable.
Target label	Low, moderate or high pollution risk	The category the model learns to predict.

The target labels must be defined consistently. For example, low risk may mean no notable pollution evidence, moderate risk may mean closer review is needed, and high risk may mean the observation meets a documented alert threshold. Environmental specialists and local standards should define the thresholds; they should not be chosen only to make the model appear accurate.

4.3 Data representation

For tabular sensor data, each row represents one sampling event at a particular place and time, while columns represent measured features and the target label. Satellite or drone imagery is represented as pixels or extracted image features. If data from different sources are combined, they need matching time and location references so that the model does not compare unrelated measurements.

5. Data Collection and Preparation

A reliable model depends on the quality and coverage of its data. Possible sources include coastal water-quality monitoring stations, field sampling records, public environmental datasets, satellite observations, weather services and verified pollution incident reports. Each record should include its source, collection time, measurement units and how its label was assigned.

5.1 Data preparation steps

1. Remove duplicate records and check whether dates, locations and measurement units are valid.
2. Inspect missing values. Use a documented method or exclude a row when necessary; do not silently replace every missing value with zero.
3. Check outliers against field notes and instrument limits. A genuine pollution event may be an extreme value, so outliers should not be removed automatically.
4. Convert categories into a format the algorithm can use and standardize units across data sources.
5. Scale numerical features if the selected algorithm benefits from it. Tree-based models generally do not require feature scaling.
6. Check class balance. If high-pollution cases are rare, consider class weighting or careful resampling on the training data only.
7. Split data into training and testing sets. For observations linked by location or time, a location-based or time-based split can reduce information leakage.

Data leakage is an important risk. If records from the same pollution incident appear in both training and testing sets, the test score may look better than real-world performance. A more realistic test keeps later dates or separate locations aside until final evaluation.

6. Analysis Using NumPy and pandas

NumPy and pandas can inspect, clean and summarize a tabular dataset before training. pandas stores data in a DataFrame and supports missing-value checks, grouping and descriptive statistics. NumPy supports numerical calculations and arrays. The following code uses illustrative values to demonstrate the analysis process; it is not a report of real field measurements.

```
import pandas as pd
import numpy as np

# Illustrative example only; replace with verified monitoring data
data = {
    'turbidity': [4.2, 8.1, 16.5, 11.3, 3.8, 22.4],
    'dissolved_oxygen': [7.1, 6.5, 4.2, 5.1, 7.4, 3.7],
    'pH': [8.1, 8.0, 7.8, 7.9, 8.2, 7.6],
    'temperature': [27.1, 28.0, 29.2, 28.7, 26.9, 30.1],
    'pollution_risk': ['Low', 'Low', 'High', 'Moderate', 'Low', 'High']
}
df = pd.DataFrame(data)
```

```
print(df.head())
print(df.isnull().sum())
print(df.describe(numeric_only=True))
print(df['pollution_risk'].value_counts())
print('Mean turbidity:', np.mean(df['turbidity']))
print(df.groupby('pollution_risk')[[
    'turbidity', 'dissolved_oxygen'
]].mean())
```

In a real dataset, these summaries help answer questions such as: Are high-risk labels associated with unusual turbidity or lower dissolved oxygen? Are some coastal locations repeatedly flagged? Are missing measurements likely to bias the result? These patterns are clues, not proof of causation. The model must be evaluated on separate observations before it supports decisions.

7. Selection of Machine Learning Techniques

7.1 Decision Tree classification

A Decision Tree is a suitable first model because it handles tabular environmental features and expresses decisions as a series of understandable tests. For example, a tree may examine turbidity, then dissolved oxygen, and then rainfall or location context before assigning a risk category. This is only an illustration; the actual splits must be learned from labelled data.

Decision Trees choose splits that improve separation between classes. Common criteria include Gini impurity and information gain based on entropy. A tree that grows too deep can memorize the training data, so maximum depth, minimum samples per leaf and cross-validation can help control overfitting. A compact tree can help environmental staff understand which features influenced a prediction.

7.2 Random Forest as a comparison model

A Random Forest combines predictions from many decision trees trained on varied samples and feature subsets. It can generalize better than a single tree when relationships are complicated. However, its overall reasoning is less direct to explain than a small tree. Comparing both models is useful: the Decision Tree provides interpretability, while the Random Forest provides a stronger benchmark.

7.3 Connection to Unit I concepts

Concept Learning and Version Spaces: A simplified version of the task can define a concept such as “high pollution risk” using conditions on measured features. Candidate-Elimination represents hypotheses that remain consistent with labelled examples. In a real marine dataset, measurement noise and complex interactions make a simple version-space model less practical than a decision tree, but the concept-learning idea explains how examples guide a classifier.

Inductive Bias: Every learning algorithm has assumptions about the patterns it can learn. A Decision Tree prefers rules formed by feature-based splits, while a Random Forest combines many such trees. Predictions can be affected by the training locations, seasons, labels and selected features. The model should be tested on new conditions rather than assumed to work equally well in every ocean region.

Heuristic Search: During tree construction, the algorithm evaluates candidate splits and chooses one that improves a criterion such as information gain. This is a heuristic search through possible decision rules. It is efficient, but a locally good split does not guarantee a perfect final tree.

8. Proposed System Workflow

1. Collect water sensor readings, verified sample results, weather information and suitable satellite or drone observations.
2. Clean and align records by location and time; document missing values and measurement units.
3. Create labels from reliable laboratory results, expert assessment or official incident records.
4. Explore the dataset with pandas and NumPy, then visualize distributions and possible relationships.
5. Train a Decision Tree and a Random Forest using the training data.
6. Evaluate both models on held-out locations or dates using precision, recall, F1-score and a confusion matrix.
7. Display risk categories and uncertainty on a map or dashboard. Send high-risk alerts for human review.
8. Verify alerts through field sampling, add confirmed outcomes to the dataset and retrain the model periodically.

The system should show the date and location of each prediction, the risk category, important measurements and an uncertainty indicator where available. A high-risk prediction should trigger an inspection recommendation, not an automatic public claim that a particular organization or community caused pollution.

9. Visualization and Interpretation

Visualization makes results easier to interpret and can reveal data problems that are difficult to see in a table. A monitoring dashboard could include:

- A coastal map showing sampling locations and risk categories, with a legend and date filter.
- A bar chart of low-, moderate- and high-risk observations to reveal class imbalance.
- A line chart of turbidity, dissolved oxygen or other measurements over time at each station.
- A scatter plot comparing turbidity with dissolved oxygen, coloured by the verified risk label.
- A confusion matrix comparing predicted categories with known test labels.

Interpretation must remain careful. If high-risk samples show higher turbidity, this may be a useful association, but rainfall, sediment disturbance or sampling conditions could also explain it. A location with many alerts may simply have more monitoring stations. Charts should include sampling effort and time period whenever possible.

10. Model Evaluation

Accuracy alone is not enough for marine pollution monitoring. If most observations are low risk, a model could achieve high accuracy by predicting low risk almost every time and still miss dangerous events. Evaluation should include class-wise precision, recall and F1-score.

Measure	Meaning for this application
Accuracy	Share of all test observations classified correctly.
Precision for high risk	Of observations flagged high risk, how many were actually high risk? Low precision can cause unnecessary inspections.
Recall for high risk	Of actual high-risk observations, how many did the model detect? Low recall means events may be missed.
F1-score	Balances precision and recall for a class.
Confusion matrix	Shows which categories are confused, such as moderate risk incorrectly labelled low.

The preferred balance depends on the consequences of errors. For early warning, missing a serious event may be more harmful than sending a team to inspect a false alarm, so recall for high-risk cases may receive extra attention. Still, frequent false alarms can waste limited resources. Thresholds should be selected with environmental experts and tested on data from different seasons and locations.

11. Expected Outcomes and Practical Applications

If the system is trained on sufficiently accurate and representative data, it may help teams identify patterns that require investigation, prioritize sampling locations and track changes over time. Possible applications include:

- Early warning for unusual water-quality readings near river mouths, ports or industrial coastlines.
- Prioritization of field sampling after heavy rainfall or a reported spill.
- Monitoring recurring pollution hotspots and checking whether conditions improve after intervention.
- Combining remote sensing with local sensor readings to broaden coverage between sampling visits.
- Providing authorities and researchers with a consistent summary of environmental observations.

No numerical improvement or model accuracy is claimed here because no real dataset has been supplied or tested. The expected benefit must be confirmed through a pilot study with verified observations, independent evaluation and feedback from monitoring teams.

12. Advantages and Limitations

Advantages

- Can analyse many observations faster than manual review alone.
- Can combine data sources to reveal patterns across time and location.
- May direct limited field-testing resources to suspicious areas.

- Decision Tree outputs can be presented as understandable rules.
- Repeated monitoring can help track trends and evaluate response activities.

Limitations

- Reliable labels are expensive and require laboratory tests or expert verification.
- Sensors may drift, fail or record readings affected by weather and natural conditions.
- Satellite imagery may miss submerged, small or cloud-covered pollution and may not identify a pollutant with certainty.
- High-pollution cases can be rare, making the model prone to missing them.
- A model trained in one coastal region may not transfer well to another region.
- Incorrect alerts can waste resources or lead to unfair blame if treated as proof.
- ML cannot remove pollutants; it must be paired with prevention, regulation, cleanup and community action.

13. Recommendations

- Start with a small pilot at selected coastal monitoring stations and expand only after evaluation.
- Use verified laboratory results and document how each pollution label was assigned.
- Include seasonal, weather and location context to reduce confusion between natural events and pollution.
- Test the model on separate dates and locations, not only random rows from the same site.
- Report precision and recall for high-risk cases alongside accuracy, and review false negatives carefully.
- Keep an expert-in-the-loop process: predictions should prompt inspection and confirmation rather than automatic enforcement.
- Protect sensitive location data where necessary and make the dashboard accessible to local authorities and researchers.
- Share standardized, quality-controlled observations among research groups and agencies where appropriate.
- Connect alerts to practical responses such as sampling, source investigation, cleanup and public communication.

14. Conclusion

Machine Learning can strengthen marine pollution monitoring by analysing water-quality measurements, location and time information, weather conditions and suitable remote-sensing data. A supervised classification model can estimate whether an observation falls into a low-, moderate- or high-risk category. A Decision Tree is a useful starting point because its rules can be inspected, while a Random Forest provides a comparison for potentially stronger predictive performance.

The reliability of the system depends on representative data, careful preprocessing, trustworthy labels and evaluation on new locations or time periods. Charts and maps help people understand patterns, but predictions must be checked through field investigation and laboratory testing. This approach supports SDG 14 – Life Below Water by helping identify possible pollution hotspots and prioritize action. Its

real value comes when accurate monitoring is combined with prevention, accountability and protection of marine ecosystems.

References

1. United Nations. (n.d.). Goal 14: Conserve and sustainably use the oceans, seas and marine resources. https://sdgs.un.org/goals/goal14

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5. scikit-learn developers. Decision Trees and Random Forests: User Guide. https://scikit-learn.org/stable/supervised_learning.html

6. pandas documentation. https://pandas.pydata.org/docs/

7. NumPy documentation. https://numpy.org/doc/

Note on data and results: The small values in the code example are illustrative only. Replace them with a documented dataset before reporting measured model performance or drawing site-specific conclusions.